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Abstract: To fully grasp the corporate information disclosure cycle, one needs to study both greenwashing and brownwashing (the shrouding of negative information).

While greenwashing research has grown substantially, strategic underreporting (brownwashing) has received less attention.

This study investigates the combined effect of voluntary sustainability-standard adoption and internal governance on brownwashing by publicly listed firms in the five largest ASEAN economies (ASEAN-5) from 2014 to 2024.

Using panel models with firm- and year-fixed effects on 1,506 firm-year observations from 385 listed companies, we examine brownwashing across both the extensive margin (decision to report) and intensive margin (degree of underreporting).

The adoption of the Global Reporting Initiative (GRI) framework decreases the likelihood of brownwashing. However, once brownwashing occurs, estimates indicate a 13.37% increase in underreporting severity.

This extensive-margin effect is more precisely estimated than the intensive-margin effect.

Across robustness checks, the directional pattern holds, though statistical significance is weaker in the smallest subsamples.

No direct evidence connects board gender diversity or board sustainability committees to brownwashing, notably in an environment where female directors occupy an average of only 18.97% of seats.

Sustainability committees also show no association with GRI adoption, indicating that external reporting guidelines and internal governance structures act as independent channels.

These findings broaden the scope of ESG decoupling research and suggest that regulators in developing economies should pair voluntary reporting standards with independent assurance rather than relying solely on board governance.

Keywords: ASEAN-5; Board gender diversity; Brownwashing; ESG decoupling; Global Reporting Initiative (GRI); Sustainability committee.

Contribution/Originality: This study investigates whether the adoption of GRI frameworks impacts the level of corporate brownwashing in two margins: extensive and intensive, and whether external GRI and internal governance mechanisms act as substitutes or complements.

Findings show no significant relationship.

Sustainability committee presence is not a significant determinant of GRI adoption.

Neither sustainability committees nor board gender diversity are significant determinants of corporate brownwashing.

The main implication is that these governance mechanisms operate through distinct channels.

As societal and regulatory pressures increase, firms rely more on non-financial disclosures to acquire legitimacy (Eliwa et al., 2023).

Effective sustainability practices can yield direct market benefits including reduced cost of capital and increased firm value.

However, the challenge is that sustainability reporting is voluntary and not directly verifiable.

This creates information asymmetry between corporate insiders and external parties (Roszkowska-Menkes et al., 2024), leaving room for strategic disclosure where there is a gap between firms' words and their actual practices. We call this misalignment between sustainability reporting and real firm behavior ESG (environmental, social and governance) information decoupling.

It can take two forms: firms exaggerating their environmental ethics (greenwashing) or firms minimizing their sustainability achievements (brownwashing).

Initial perspectives treated decoupling as unidirectional, mainly focusing on low sustainability firms overstating their ethical commitment.

But Kim and Lyon (2015) challenged this narrow view: decoupling is bidirectional, with firms brownwashing by understating genuine sustainability gains they have achieved.

Firms employ these strategies to shape stakeholder expectations and avoid the operational and reputational costs of setting performance goals they may fail to meet.

The literature on decoupling has three distinct gaps.

First, the thematic gap: current research focuses heavily on greenwashing with less attention to brownwashing. Second, the structural gap: research on external drivers like the adoption of Global Reporting Initiative (GRI) standards and internal drivers like having diverse boards or sustainability committees too often occurs in silos, with each driver viewed as one of many in a system of independent effects.

Third, the geographic gap: empirical evidence on decoupling remains concentrated in Western and developed markets, leaving limited insight into fast-growing ASEAN-5 (the five largest economies of the Association of Southeast Asian Nations), which are attracting ESG investments in massive quantities.

The heterogeneity of their disclosure frameworks is stark: Singapore (Singapore Exchange, SGX) has mandated ESG disclosure earlier than its regional peers, while others still have voluntary or transitional regimes.

This paper examines the roles of reporting standards and internal governance mechanisms in ASEAN-5 economies (Malaysia, Singapore, Thailand, Indonesia, and the Philippines) on corporate brownwashing from 2014 to 2024.

ASEAN-5 is an interesting context for studying the drivers of bidirectional decoupling given the diversity in regulatory governance and the scale of the economies.

Drawing on four theories, namely neoinstitutional, stakeholder-agency, gender socialization, and resource dependence theory, we formulate and test four hypotheses.

First, GRI adoption reduces corporate brownwashing (H1).

Second, greater board gender diversity lowers the extent of corporate underreporting (H2).

Third, existence of sustainability committee (H3) directly lowers brownwashing.

Fourth, sustainability committee existence increases the likelihood of GRI adoption (H4).

Results from the fixed-effects panel analyses reveal a narrower pattern of relationships than hypothesized.

Adoption of the GRI reduces brownwashing on both the extensive and intensive margins, with a more precisely estimated effect at the extensive margin.

However, corporate board gender diversity and the existence of sustainability committees show no significant relationship with brownwashing.

The paper makes three related contributions to the literature on disclosure.

First, from a conceptual point of view, it contributes to the literature by refining the notion of greenwashing through decomposing information decoupling into positive and negative exclusive parts and studying the latter in isolation.

Second, it studies the interaction between external disclosure standardization and internal governance mechanisms.

The evidence suggests the absence of a reinforcing effect: the presence of sustainability committees does not facilitate the adoption of GRI either as a substitute or a catalyst.

Third, it distinguishes the extensive from the intensive margin of underreporting, showing that standardized frameworks reduce both the likelihood that brownwashing starts and its severity once underway, with effects similar in magnitude but more precisely estimated for prevention than for cure.

The paper proceeds as follows: Section 2: Literature Review; Section 3: Theoretical background and Hypotheses, Section 4: Data and Methodology, Section 5: Results and Discussion, and Section 6: Conclusions.

Strong ESG disclosure leads to increased legitimacy and better access to capital, reflected in lower cost of debt and higher valuations (Eliwa et al., 2023; Saeed et al., 2025).

This presents a tangible incentive for companies to disclose strong sustainability performance.

However, sustainability reporting is mostly voluntary and hard to verify (Roszkowska-Menkes et al., 2024).

This leads to a situation where stated practices may diverge from actual conduct.

Some companies portray inflated image of ethical conduct that does not match their practices (Cepêda, 2026; MacLean & Behnam, 2010).

In the short term this may boost the reputation but once uncovered it results in market penalties (Di & Li, 2023; Eliwa et al., 2023).

The concept of decoupling stems from the neoinstitutional theory where organizations adopt formal policies to symbolically comply without changing their actual practices (Meyer & Rowan, 1977; Roszkowska-Menkes et al., 2024).

Kim and Lyon (2015) expanded this into a bidirectional framework where organizations seeking to avoid scrutiny may either overstate their positive actions (greenwashing) or understate their genuine achievements (brownwashing).

Recent literature begins to operationalize the conceptualization of a reporting-performance gap in ESG practices. Hawn and Ioannou (2016) pioneered empirics by contrasting nominal ESG scores with controversy-adjusted scores.

Building on that work, an emerging stream of research conceptualizes decoupling as bidirectional rather than a unidirectional construct (Moreno-Ureba, 2026; Yue et al., 2026).

Prior studies identify external and internal forces that limit decoupling.

Externally, standardized reporting frameworks and assurance serve as limits to symbolic ESG practices (García-Sánchez et al., 2022).

Yue et al.

(2026) identify an internal limit: the positive feedback of ESG performance can increase stakeholders' expectations of CSR such that brownwashing is a means to alleviate scrutiny.

Eliwa et al.

(2023) show that female board representation correlates with a lower degree of ESG decoupling, and the presence of sustainability committees (Qiu et al., 2026) and specialized board expertise (Moreno-Ureba, 2026) shape ESG reporting.

Conceptualizations of bidirectional decoupling remain vague in our field, with most concentrating on greenwashing by itself (Moreno-Ureba, 2026).

Likewise, there are only a few explorations of the joint effect of external forces such as GRI adoption and internal governance mechanisms.

Moreno-Ureba (2026) calls for research that disaggregates decoupling and tests the moderating role of internal governance; this study takes up that call by analyzing GRI adoption and internal governance together.

Global standardization of sustainability reporting provides a key external guardrail to opportunistic disclosure. Standardized non-financial metrics built around structured and uniform reporting frameworks enable cross-corporate comparability (Friske, 2024; Sannino et al., 2020).

Over the last two decades, standards have gradually incorporated more multi-dimensional indicators into their templates, resulting in an increase in disclosure quality and a decrease in corporate opaqueness (García-Sánchez et al., 2022; Roszkowska-Menkes et al., 2024).

There are two competing views on how uniform reporting influences strategic disclosure: as a genuine information-discipline mechanism, or as an opportunity for narrative camouflage, developed formally in Section 3.2.

Our evidence indicates that in the ASEAN-5 countries the key voluntary reporting framework is the Global Reporting Initiative (GRI), adopted by 85.16% of firms.

The aggregate figure masks cross-country heterogeneity in the pace of adoption and quality of compliance with standards.

Institutional paths diverged, with Singapore pushing mandatory general ESG disclosure early on, though the choice to report specifically under GRI remains voluntary even there, as Malaysia, Thailand, Indonesia, and Philippine firms staggered through multiple phases of voluntary guidance and new frameworks.

This heterogeneity manifests in an uneven spread of standardized reporting during 2014-2024, with adoption driven by institutional pressure in home countries (García-Sánchez et al., 2022; Sannino et al., 2020).

It was founded to harmonize reporting of non-financial information, but in the context of ASEAN-5, with uneven enforcement and institutional differences, firms will still diverge.

Narrative camouflage will persist in weak-oversight regions via selective disclosure to highlight positive aspects and to obscure negative ones.

Stakeholder theory is the underpinning theory since stakeholders are "any group or individual who can affect or is affected by the achievement of the organization's objectives" (Freeman, 1984, p. 46), and the stakeholder groups provide resources a firm needs to survive and grow and demand accountability (Roszkowska-Menkes et al., 2024).

Note that stakeholders hold misaligned interests.

For example, community groups may seek more environmental disclosure whereas shareholders may worry that such disclosure jeopardizes short-term profits.

Executives, as agents of the firm, must manage these competing demands (Qiu et al., 2026).

Stakeholder-agency theory helps explain this behavior by arguing that differences in the sources of power and information asymmetry influence which demands managers respond to, and the internal governance structures constrain opportunistic disclosure that aligns manager behavior with stakeholder expectations (Moreno-Ureba, 2026).

Information discipline, the pressure that standardized, comparable metrics place on managers to align disclosed and actual practice, should improve transparency when firms adopt frameworks such as the GRI, which equips firms with a vocabulary for rigorous and comparable disclosure (Friske, 2024; Wang et al., 2026).

Information discipline is the mechanism through which GRI adoption should reduce brownwashing.

Standardized data collection and reporting across multiple ESG indicators makes firm performance easier to evaluate by rating agencies and analysts and so increases the visibility cost of what Kim and Lyon (2015) term "green hushing," the informal label for the same brownwashing behavior this study measures.

This in turn should create pressure on managers to narrow the negative decoupling gap by aligning actual conduct with disclosed commitments.

However, as mentioned before, a less optimistic view is narrative camouflage.

Since voluntary guidelines such as GRI allow flexible application symbolic compliance remains possible (Roszkowska-Menkes et al., 2024).

Because sustainability reporting is voluntary and unverifiable, insiders know more about actual practices than outside stakeholders can observe, and managers might use standardized metrics as a "ceremonial shield" to

draw attention to favorable data points and obscure unfavorable ones in lengthy qualitative disclosures (Boiral et al., 2019; Wijen, 2014).

From this perspective voluntary standards may not function as a binding constraint on communication and could even enable misrepresentation rather than transparency (Roszkowska-Menkes et al., 2024).

Section 5 tests which of these two effects, information discipline or narrative camouflage, dominates empirically.

Hypothesis 1 (H1).

GRI reporting-standard adoption correlates negatively with corporate brownwashing.

The literature widely accepts the link between board gender diversity and firm transparency (Eliwa et al., 2023; Moreno-Ureba, 2026).

Gender socialization and stakeholder-agency theories provide the major foundation for this link.

These theories suggest that male and female directors have distinct cognitive and ethical frameworks and that they vary in their tolerance of risk because of their different socialization (Boulouta, 2013; Manita et al., 2018).

This division has ramifications for a board's oversight capacity in practice.

Research indicates that female directors have a greater orientation toward stakeholders and a greater sensitivity to social and environmental concerns (Eagly et al., 2003; Eliwa et al., 2023).

Gender diverse boards, then, should be more vigilant monitors, less likely to accept unethical or deceptive firm disclosure (Manita et al., 2018) and less likely to excuse managers for misreporting.

The opposing perspective, though, is a more guarded one.

Adding women to the board may not produce effective governance, since the dynamics of inequality and tokenism can diminish their influence (Kanter, 1977; Konrad et al., 2008; Manita et al., 2018).

Board composition alone is not sufficient; the level of influence women hold within the board matters as much.

Healthy male majority on board may dilute board's ethical voice because prevailing norms absorb or suppress that voice (Torchia et al., 2011).

Again, female directors may not have enough tenure or authority to influence decisions (Moreno-Ureba, 2026).

Practically, extending disclosure requires less organizational will than changing core practices.

Thus, female directors might affect reporting practices more than core practices, decreasing brownwashing when performance is strong but increasing greenwashing when performance is weak.

Even so, female directors' monitoring role should still deter non-disclosure overall, supporting a negative sign for H2.

Hypothesis 2 (H2) follows: board gender diversity correlates negatively with corporate brownwashing.

It is important to study external voluntary standards and internal corporate governance jointly when explaining strategic disclosure (Qiu et al., 2026).

Resource dependence theory and stakeholder-agency theory suggest internal governance, especially sustainability committees, works with GRI adoption as a complementary control.

Board-level sustainability committees should improve the oversight of ESG risks and reporting.

This is relevant for brownwashing and reporting as well.

First, a board-level sustainability committee is a subgroup of the board of directors dedicated to handling ESG risks and reporting (Qiu et al., 2026).

Structurally, sustainability committees provide the board of directors with additional capacity and expertise, which general boards lack (Qiu et al., 2026; Velte, 2017).
In theory and practice, then, sustainability committees serve as more rigorous internal monitors (Velte, 2017).

Stakeholder-agency theory provides another perspective.

An active sustainability committee can help firms achieve substantive ESG-related goals instead of symbolic gestures (Ștefănescu & Suciu, 2026).

This reduces the likelihood of brownwashing through consistent communication with stakeholders and setting concrete ESG targets.

Based on this we propose the third hypothesis: Hypothesis 3 (H3).

A board-level sustainability committee correlates negatively with corporate brownwashing.

Second, the board-level sustainability committee's role in ESG reporting extends beyond direct oversight.

The committee can act as a catalyst for more formalized ESG reporting.

Resource dependence theory suggests that the presence of a board-level sustainability committee predicts a firm's willingness to adopt GRI reporting guidelines.

Covering as many indicator categories as GRI does requires specialized expertise to collect and disclose multi-dimensional data.

A sustainability committee helps meet that requirement by bringing directors with relevant experience onto the board (Qiu et al., 2026), and by keeping non-financial accountability on the agenda, it pushes the board toward formal, indicator-based reporting.

We therefore expect listed firms with such a committee to be more likely to adopt GRI standards:

Hypothesis 4 (H4).

The presence of a board-level sustainability committee correlates positively with the likelihood of GRI adoption.

Figure 1 depicts the framework of our research.

Figure 1.

Proposed conceptual framework.

Source: Authors' own work.

We collect data from multiple databases using ISIN (International Securities Identification Number) and fiscal year as the matching key (Eliwa et al., 2023).

We drop observations whose firm-years do not match across databases or that have incomplete data, then apply the further filters described below.

We obtain the 2014–2024 ASEAN-5 sample from three databases: we derive ESG data from LSEG (London Stock Exchange Group) Workspace, obtain governance data from BoardEx and official corporate disclosures, and extract financial data from Compustat Global.

The estimation sample consists of 40,799 firm-years observed over 3,709 firms.

We exclude observations with missing values, financial firms, and firms not reporting CO2 emissions, which leaves 1,995 firm-years that satisfy the data-availability filters.

We handle remaining missing values through listwise deletion and chained-equations imputation.

We lag all independent, control, and moderating variables (FemD, BSize, InD, SustCom, Size, Lev, ROA, MTB, GRI, log_co2; defined in Table 2 below) by one year (t-1) relative to the brownwashing outcome (t); CM, Blau, and Shannon, described below, serve as descriptive-only alternatives to FemD, and we do not enter them as regressors.

We use this lag structure to reduce concerns about reverse causality and to mirror the realistic time lag in the effect of governance and reporting practices on corporate disclosures.

The lag structure removes each firm's first sample year from estimation, which leaves an estimation panel of 1,506 firm-years over 385 firms (see Table 1).

Table 1.

Sample distribution by country

Country

Firms

Firm years

Malaysia (Bursa)

132

459

30 5

Thailand (SET)

113

435

28 9

Singapore (SGX)

64

299

19 9

Indonesia (IDX)

47

179

11 9

Philippines (PSE)

29

134

8 9

Total

385

1 506

100

Note: Bursa = Bursa Malaysia; SET = Stock Exchange of Thailand; SGX = Singapore Exchange; IDX = Indonesia Stock Exchange; PSE = Philippine Stock Exchange.

Source: Authors' own work.

We winsorize ROA, Lev, MTB, and log_co2 at the 1st and 99th percentiles to reduce the influence of outliers, and retain their un-winsorized versions for unreported robustness checks

Table 2.
Variable definitions and measurement

Variable

Detailed Definition and Measurement

Panel A: ESG information decoupling indices

GWl

Normalized greenwashing gap ($z\text{-score}(\text{ESG_Score}) - z\text{-score}(\text{ESGC_Score})$).

Original variable; we decompose GREENWASH and BROWNWASH directly from it.

Hawn & Ioannou (2016); Moreno-Ureba (2026); García-Sánchez et al. (2022); MacLean & Behnam (2010); Di & Li (2023)

GREENWASH

Positive part of GWI: = GWI if $\text{GWI} > 0$, otherwise = 0.

Kim & Lyon (2015); Moreno-Ureba (2026); Boiral et al. (2019); Crilly et al. 2016

BROWNWASH

Negative part (absolute value) of GWI: = $-\text{GWI}$ if $\text{GWI} < 0$, otherwise = 0.

Kim & Lyon (2015); Moreno-Ureba (2026); Yue et al. (2026); MacLean & Behnam (2010); Wijen (2014)

Algebraic constraint: $\text{GWI} = \text{GREENWASH} - \text{BROWNWASH}$; the two variables are mutually exclusive. Frequencies sum to 100%, confirming exclusivity.

Nominal ESG score (voluntary information).

Nominal ESG score voluntary information.

(2023); Moreno-Ureba (2026); Roszkowska-Menkes et al. 2023 Moreno Ureba 2026 Roszkowska Menkes et al. 2024

ESGC Score

Practical ESG score controversy adjusted.

(2023); Moreno-Ureba (2026); Roszkowska-Menkes et al. 2023 Moreno Ureba 2026 Roszkowska Menkes et al. 2024 Di Li 2023

Panel B Independent control and moderating variables

FemD

of female board members.

(2023); Moreno-Ureba (2026); Manita et al.
(2018); Boulouta (2013); Konrad et al.
2018 Boulouta 2013 Konrad et al.
2008

CM

Critical mass 1 if female 30.

Kanter 1977 Torchia et al.
2011

Blau diversity index (from FemD).

Blau diversity index from FemD.

Eliwa et al.
2023 Moreno Ureba 2026

Shannon entropy index (from FemD).

Shannon entropy index from FemD.

Eliwa et al.
2023 Moreno Ureba 2026

BSize

Board size total members.

Eliwa et al.
2023 Moreno Ureba 2026

InD

Board independence.

(2023); Moreno-Ureba (2026); Manita et al.
2023 Moreno Ureba 2026 Manita et al.
2018

SustCom

Sustainability committee 1 if present.

Qiu et al.

2026 Velte 2017 Stefanescu Suciú 2026

Size

Natural logarithm of total assets.

Eliwa et al.

2023 Moreno Ureba 2026 Cepeda 2026

Lev

Financial leverage.

(2023); Moreno-Ureba (2026); Saeed et al.

2023 Moreno Ureba 2026 Saeed et al.

2025

ROA

Return on total assets.

(2023); Moreno-Ureba (2026); Saeed et al.

2023 Moreno Ureba 2026 Saeed et al.

2025

MTB

Growth opportunities Market Book value.

(2023); Moreno-Ureba (2026); Saeed et al.

2023 Moreno Ureba 2026 Saeed et al.

2025

GRI

GRI reporting standard adoption 1 if adopted.

(2026); Roszkowska-Menkes et al.

(2024); García-Sánchez et al.

(2022); Sannino et al.

2022 Sannino et al.
2020

Log of actual CO2 emissions.

Log of actual CO2 emissions.

Moreno Ureba 2026 Yue et al.
2026

Note This table defines the study s key variables their measurement and the sources that motivate their inclusion. Panel A lists the ESG information decoupling indices Panel B lists the independent control and moderating variables.
Section 4 explains why we report CM Blau and Shannon here as descriptive alternatives to FemD but exclude them from the regression tables.

Source Authors own work.

Table 2 also reports three alternative measures of gender diversity critical mass CM the Blau index and the Shannon entropy index.
We include these for completeness and for comparability with prior studies.
Because $Blau = 1 - \sum p_i^2$ and $Shannon = -\sum p_i \ln p_i$ where p_i is the share of female directors $FemD = 100$ and $q = 1 - p$ both indices are deterministic monotonic transformations of FemD and CM is a binary recoding of the same underlying variable at the 30 threshold Kanter 1977.

Entering FemD alongside any of these transformations would introduce near perfect collinearity rather than new information so we retain FemD as the continuous gender diversity measure in Tables 4 and 5 since it uses the full information in board composition rather than a threshold or a rescaled aggregate.
CM Blau and Shannon appear in Tables 2 and 3 for descriptive completeness only.

Table 3.
Descriptive statistics

Variable

Mean

Std.
Dev.

Min

Median

Max

Frequency

Panel A ESG information decoupling indices

GW

0 00

1 47

3 43

0 14

8 40

GREENWASH

0 51

1 08

0 00

0 00

8 40

45 56

BROWNWASH

0 51

0 68

0 00

0 14

3 43

54 44

ESG Score

55 30

15 66

5 34

55 95

92 07

ESGC Score

97 13

11 89

4 55

100 00

100 00

Panel B: Independent, control, and moderating variables

FemD

18 97

13 20

0 00

16 67

75 00

CM

19 95

Blau

0 27

0 15

0 00

0 28

0 50

Shannon

0 42

0 21

0 00

0 45

0 69

BSize

9 54

3 15

1 00

9 00

21 00

lnD

49 37

14 36

0 00

50 00

100 00

SustCom

22 71

Size

21 62

1 55

16 31

21 79

25 34

Lev

28 77

17 14

0 15

29 64

66 37

ROA

5 84

6 40

9 12

4 40

36 84

MTB

3 00

5 48

0 24

1 42

42 01

GRI

85 16

log co2

11 77

2 66

2 38

11 64

19 10

Note This table reports descriptive statistics for the study s key variables mean standard deviation minimum median and maximum for continuous variables and frequency for binary variables.

Source Authors own work.

We test the hypotheses with panel regressions that include firm fixed effects u_i which absorb time invariant unobserved heterogeneity and year fixed effects l_t which absorb macroeconomic shocks and region wide regulatory trends common to all firms in a given year.

We cluster standard errors at the firm level throughout Wooldridge 2010.

To test the baseline effects of GRI FemD and SustCom on brownwashing H1 H3 we estimate

where i and t denote firm and year Controls it includes BSize lnD Size Lev ROA MTB and log co2 and e_{it} is the error term.

Because 45 56 of the sample has a BROWNWASH value of zero Table 3 we define the measure formally below Ordinary least squares OLS estimates of Model 1 may not be consistent.

Following standard corner solution treatment Wooldridge 2010 we therefore split the analysis into an extensive margin whether a firm brownwashes at all and an intensive margin how much it brownwashes conditional on brownwashing.

We model the extensive margin with a linear probability model LPM that includes firm and year fixed effects

In this specification the GRI coefficient is directly interpretable as a percentage point change in the probability of brownwashing.

We estimate the intensive margin underreporting severity conditional on $GW_i = 0$ GW_i is the continuous decoupling gap defined in Table 2 BROWNWASH equals GW_i when GW_i is negative and 0 otherwise using Eq. 1 on the subsample of firm years that brownwash.

As a supplementary test of H3's logic we also ask whether internal governance moderates the effect of external reporting pressure on the extensive margin.

We add $GRI_{it} \times SustCom_{it}$ to the linear probability model reported in Table 5 Panel A

Finally to test whether sustainability committees are an antecedent of GRI adoption H4 we estimate

As a robustness check we also estimate a Heckman two stage model Heckman 1979 to address possible sample selection since only firm years with a complete set of lagged governance and financial covariates enter the estimation panel Qiu et al 2026.

Probit model for selection performed on 1995 firm years post initial data filtrations described in Section 4 but prior to the one year lag that drops each firm's first sample year.

The objective of this first step is to predict the probability that a firm year enters the analytic panel.

The exclusion restriction is defined as a firm's leave one out GRI adoption rate i.e. the mean GRI adoption status over its other sample years.

Specifically this variable reflects how consistently a firm is observed in the analytic sample conditional on reporting propensities but does not directly affect the brownwashing decision in any particular year.

The identification strategy is validated by several robustness checks.

In particular We re estimate the extensive margin specification with two way clustered firm and year standard errors which account for the cross sectional dependence of shocks and unobservables Wooldridge 2010.

Because the firm fixed effects model exploits within firm variation to identify the coefficient of GRI We re estimate the baseline specification on narrower samples.

One is the switchers sample consisting of firms that switch GRI adoption status at least once during the sample period.

The other is the clean switchers sample consisting of firms that switch GRI status only once and do not switch it back.

A final check is to compare the baseline model with the pooled model evaluating whether the omitted variable bias is sizeable

We first estimate the baseline relationships between GRI adoption internal governance and brownwashing using firm and year fixed effects panel models.

Table 4 reports three specifications Column 1 is the pooled model for the continuous brownwashing gap Eq.

1 Column 2 is the extensive margin Eq.

2 and Column 3 is the intensive margin the same continuous specification as Eq.

1 but restricted to firm years with $GW_i = 0$.

Table 4.
The extensive and intensive margins of GRI adoption on corporate brownwashing behaviors

Variables

1

BROWNWASH

pooled

2

P BROWNWASH 1

extensive margin

3

BROWNWASH GWI 0

intensive margin

FemD

0 0011

0 0015

0 0023

0 0022

0 0018

0 0029

GRI

0 2146

0 0566

0 1337

0 0476

0 1285

0 0732

BSize

0 0075

0 0091

0 0109

0 0132

0 0043

0 0199

InD

0 0001

0 0011

0 0021

0 0016

0 0036

0 0022

SustCom

0 0202

0 0297

0 0075

0 0372

0 0376

0 0656

Size

0 0022

0 0766

0 0299

0 0798

0 2413

0 1401

ROA

0 0031

0 0023

0 0062

0 0027

0 0032

0 0050

Lev

0 0018

0 0023

0 0024

0 0022

0 0018

0 0038

MTB

0 0044

0 0024

0 0020

0 0029

0 0127

0 0054

log co2

0 0098

0 0139

0 0343

0 0185

0 0213

0 0248

Firm FE

Yes

Yes

Yes

Year FE

Yes

Yes

Yes

Observations

1 506

1 506

704

Firms

385

385

296

R squared

0 820

0 662

0 858

Note This table reports fixed effects panel regressions of GRI reporting standard adoption on brownwashing. Column 1 is the pooled model on the continuous gap column 2 is the extensive margin model whether a firm brownwashes and column 3 is the intensive margin model severity conditional on brownwashing occurring. We omit the greenwashing side results the algebraic mirror image of columns 1 2 for brevity. We report cluster robust standard errors in parentheses. denote statistical significance at 1 5 10.

Source Authors own work.

Column 1 shows that GRI adoption is significantly and negatively associated with the continuous magnitude of brownwashing ss 0 2146 p 0 01 supporting H1 Hawn Ioannou 2016.

Decomposing this relationship reveals an asymmetry.

In Column 2 GRI remains a significant negative predictor on the extensive margin ss 0 1337 p 0 01 adopting GRI lowers the probability of brownwashing by 13 37 percentage points.

Among firms that already brownwash GWI 0 Column 3 shows the GRI coefficient on the intensive margin stays negative and remains significant at the 10 level ss 0 1285 p 0 10 close in magnitude to the extensive margin coefficient.

Standardized disclosure therefore works as both a preventive and a corrective mechanism though we estimate the effect with less precision once brownwashing has already started consistent with a smaller more heterogeneous subsample of brownwashing firms.

Structured comparable GRI indicators set a transparency baseline that raises the reputational cost of brownwashing consistent with the observed reduction in its likelihood Friske 2024 Wang et al 2026 Yue et al 2026 and the voluntary loosely enforced nature of GRI adoption in emerging ASEAN markets may still explain why the intensive margin effect though present carries a larger standard error than the extensive margin effect Boiral 2013 Roszkowska Menkes et al 2024.

Board gender diversity FemD is negative but not significant in any specification ss 0 0011 p 0 10 pooled ss 0 0023 p 0 10 extensive margin so the data do not support H2.

Stakeholder agency and gender socialization theory predict that female directors act as vigilant monitors Boulouta 2013 Eagly et al 2003 Eliwa et al 2023 but our results fit a tokenism explanation better Kanter 1977 Manita et al 2018.

Female board representation in the ASEAN 5 sample averages only 18 97 Table 3 often below the critical mass needed for substantive influence Konrad et al 2008 and female directors face structural barriers to shaping capital intensive reporting strategies Moreno Ureba 2026.

Sustainability committee presence SustCom also shows no significant baseline relationship with brownwashing ss 0 0202 p 0 10 pooled ss 0 0075 p 0 10 extensive margin which does not support a direct effect.

Committees in ASEAN firms may play a symbolic legitimacy seeking role rather than exercise substantive oversight Boiral 2013 Qiu et al 2026 Stefanescu Suciu 2026 and on their own they do not appear to change disclosure incentives unless tied to an external standard such as the GRI.

Among the controls profitability ROA is negative and significant on the extensive margin ss 0 0062 p 0 05 profitable firms are less likely to brownwash which fits the behavioral theory of the firm Kim Lyon 2015 Yue et al 2026.

Profitable firms face less survival anxiety while distressed firms underreport to manage investor expectations and avoid setting a performance benchmark they cannot sustain Yue et al 2026.

Growth opportunities MTB are negative and significant on the intensive margin ss 0 0127 p 0 05 suggesting that high growth firms brownwash less severely once decoupled likely because they prioritize capital market legitimacy for external financing Cepeda 2026.

Board independence InD is positive and significant on the intensive margin ss 0 0036 p 0 10 among firms that already brownwash those with more independent boards underreport more severely a result we do not have a strong theoretical prior for and flag for future work.

Carbon intensity log co2 is negative and significant on the extensive margin ss 0 0343 p 0 10 consistent with higher emission firms facing more scrutiny and so being less likely to brownwash at all.

We run a series of econometric robustness tests to check the validity and causal identification of the baseline results.

Note: This table reports fixed-effects panel regressions of GRI reporting-standard adoption on brownwashing.

Table 5.

Advanced mechanisms structural sensitivity and endogeneity checks

Model Specification Subsample

Coefficient GRI

Standard Error

p value

N Firms

Panel A Theoretical Mechanisms and Interactions

1 Interaction with SustCom GRI x SustCom

0.1169

0.1760

0.507

1,506,385

Panel B Sensitivity of Statistical Inference

2 Full sample two way clustering Firm Year

0.1337

0.0501

0.026

1,506,385

Panel C Sensitivity of Identification Switcher Subsample

3 71 switchers only cluster Firm

0.0974

0.0512

0 062

382 71

4 71 switchers only cluster Firm Year

0 0974

0 0562

0 117

382 71

5 62 clean switchers only cluster Firm

0 0821

0 0571

0 156

302 62

Panel D Specification and Endogeneity Diagnostics

6 Industry Country Year FE no Firm FE

0 2810

0 0409

0 001

1,506 / 385

Panel E Sample Selection Correction

7 Heckman two step selection corrected

0 2799

0 0532

Note This table shows mechanism interactions sensitivity diagnostics and tests for endogeneity of the GRI effect on the extensive margin of brownwashing.

Panel A provides a supplementary test of the logic of H3 by using the interaction between GRI and sustainability committee SustCom.

Panels B C offer sensitivity diagnostics by comparing statistical power across alternative clustering and switcher subsamples.

Panel D illustrates the importance of firm fixed effects FE in purging unobserved time invariant heterogeneity and Panel E performs a Heckman two step correction for sample selection into the analytical panel reported coefficients are GRI estimates from second stage regression.

We report cluster robust standard errors in parentheses.

denote statistical significance at 1 5 10.

Source Authors own work.

The interaction term Model 1 is positive but far from significant ss 0 1169 p 0 507 so sustainability committees do not moderate the extensive margin of brownwashing.

Standardized reporting appears to act as a field level force that operates independently of internal governance GRI s comparability requirements discipline managerial communication on their own Friske 2024 Roszkowska Menkes et al 2024 leaving no additional role for committee presence.

Panel B checks whether the statistical inference holds up against cross sectional dependence and regional macroeconomic shocks.

Model 2 re estimates the extensive margin model with standard errors clustered two ways by firm and by year Wooldridge 2010.

Under this more demanding specification the GRI coefficient stays negative and significant at the 5 level ss 0 1337 p 0 026 so the disciplinary effect of standardized reporting on brownwashing holds up against more complex regional error structures.

Panel C tests the sensitivity of the identification strategy.

With the firm fixed effects specification within firm variation alone identifies the GRI coefficient drawn only from firms that change their GRI status during 2014 2024 termed switchers.

In Model 3 we restrict the sample to these 71 switchers 382 observations and cluster standard errors on firm level. The estimate of GRI remains negative and significant at 10 ss 0 0974 p 0 062.

In the following models we narrow the specification further.

In Model 4 we use two way clustered standard errors the magnitude of GRI s coefficient remains similar ss 0 0974 p 0 117 but it loses significance because of the small sample size.

Model 5 further restricts to 62 clean switchers 302 observations firms that transition to or from GRI once and do not revert to their previous status and still obtains a negative similarly sized estimate of GRI s coefficient ss 0 0821 p 0 156 though also not significant at conventional levels.

Across all three switcher specifications the point estimate stays negative and stable in magnitude consistent with within firm transitions rather than cross sectional differences between firms identifying the baseline result even though statistical power falls as the switcher subsample shrinks.

This supports the causal interpretation.

Panel D tests the sensitivity of the specification in Model 6 we remove firm fixed effects and introduce industry country and year dummies instead.

The GRI coefficient more than doubles in magnitude from 0.1337 in the firm fixed effects specification to 0.2810 p 0.001 consistent with omitted variable bias unobserved time invariant firm specific factors such as ethical culture or management quality Eliwa et al 2023 Moreno Ureba 2026 may correlate positively with the decision to adopt the GRI framework and negatively with the level of brownwashing.

Firm fixed effects are therefore necessary to isolate the effect of the institutional change in sustainability reporting practices.

In Model 7 reported in Panel E firms may have self selected into the analytic panel analytic panel entry firms may be a non random subset of all firms.

Applying Heckman two step selection correction yields the same main result the GRI coefficient remains negative and significant at 1 ss 0.2799 p 0.001 close in magnitude to the omitted variable bias estimate in Panel D Selection correction leaves the coefficient's sign and significance unchanged and the Inverse Mills ratio is itself insignificant ss 0.2916 p 0.155 full first and second stage Heckman output available from the corresponding author on request indicating that selection into the panel is not driving the result.

Because this test uses the same within firm variation as the baseline specification it should be read as a check on the baseline estimate rather than as independent evidence from a different identification source.

We next test the antecedent pathway implied by H4 whether the presence of a sustainability committee predicts a firm's future GRI adoption.

Results in Table 6 Model 1 show the antecedent test in which we regress GRI on a one year lagged SustCom in a firm and year fixed effects panel model with the same control variables used throughout the paper.

Table 6.
Antecedent test sustainability committee presence and GRI adoption H4

Model Specification

Coefficient SustCom

Standard Error

p value

N Firms

1 Firm Year FE one year lag primary specification

0.0141

0.0191

0 461

1 506 385

2 Contemporaneous SustCom no lag

0 0192

0 0177

0 279

1 506 385

3 Industry Country Year FE

0 0034

0 0219

0 878

1 506 385

4 Pooled Year FE only

0 0111

0 0218

0 611

1 506 385

Source Authors own work.

Models 2 4 vary the specification for robustness model 2 uses contemporaneous SustCom without the one year lag model 3 replaces firm fixed effects with industry country and year fixed effects and model 4 uses a pooled specification with year fixed effects only.

In all models we cluster standard errors by firm and report them in parentheses below the coefficient estimates. We include control variables board size board independence firm size leverage ROA market to book and log CO2 emissions in all specifications but omit them from the table to save space full regression coefficients are available from the corresponding author on request.

SustCom coefficient is tiny and statistically indistinguishable from zero in the main model ss 0 0141 p 0 461 meaning that the data do not support H4.

The finding is consistent across different model specifications.

We find the same null effect when we use a non lagged SustCom variable ss 0 0192 p 0 279 when replacing firm fixed effects with industry country and year fixed effects ss 0 0034 p 0 878 or when using a pooled specification with only year fixed effects ss 0 0111 p 0 611.

All these results imply that the presence of sustainability committees does not predict adoption of the GRI standards by firms.

The wider implication is that this finding challenges resource dependence theory.

External reporting and internal governance appear to work through separate channels GRI adoption is a direct discipline mechanism H1 and SustComs neither substitute for nor trigger it.

Figure 2 shows GRI adoption trends 2014 2024.

Adoption rises steadily and switching events are spread gradually across years rather than clustered around one macro shock a pattern that favors fixed effects identification because it rules out the possibility that the mitigating effect comes from a single crisis or policy reform.

Figure 2.

GRI adoption trends and switcher diffusion timing 2014 2024

Source Authors own work.

Figure 3 reports the GRI coefficient across the seven specifications in Table 5.

The coefficient is negative and stable across every specification except the GRI x SustCom interaction term which captures a different quantity the incremental effect of committee presence not the main GRI effect indicating that the mitigating effect does not depend on any particular set of controls or clustering choices.

Figure 3.

Robustness diagnostics and specification testing for the primary GRI coefficient.

Source Authors own work.

Main results focus on two way decoupling of ESG reporting practices in ASEAN 5 markets during 2014 2024 by jointly considering the effects of external reporting guidance and internal board monitoring on corporate brownwashing.

Our panel fixed effects findings indicate a consistent pattern.

The adoption of GRI sustainability reporting standards is negatively and significantly associated with brownwashing but in a non homogeneous manner.

GRI standards restrict both the extensive margin of brownwashing decisions i e whether a firm initiates brownwashing and the intensive margin i e the severity of ESG score underreporting among brownwashers with coefficients similar in magnitude but the extensive margin effect estimated with greater statistical precision.

This pattern is congruent with the information discipline hypothesis H1.

Uniform comparable indicators are likely to increase the reputational risks associated with brownwashing Garcia Sanchez et al 2022 Kim Lyon 2015 and act as a deterrent against the initiation of brownwashing practices and with

less precision against its continuation once underway though this effect is more muted for voluntary standards practiced in the developing ASEAN markets Boiral 2013 Roszkowska Menkes et al 2024.

Second neither FemD nor SustCom correlates significantly with brownwashing H2 H3 which runs against stakeholder agency and gender socialization theories but fits a tokenism explanation instead Kanter 1977 Manita et al 2018 women still hold only 18.97% of ASEAN boardroom seats on average short of the critical mass needed to affect disclosure decisions Moreno Ureba 2026 and sustainability committees may function as a legitimacy seeking symbol rather than a substantive monitor Qiu et al 2026.

Third the antecedent model results indicate that sustainability committee presence is not a significant predictor of the adoption of GRI guidelines.

The coefficient estimates for this variable in all model specifications tested are close to zero and statistically insignificant see Table 6.

So not support for H4 These results shift the analytical focus from the well studied unidirectional exaggeration of greenwashing to the more complex intentional underreporting of brownwashing answering Moreno Ureba's 2026 call to disentangle corporate decoupling.

With this we move the analytical framing away from unidirectional exaggeration greenwashing and into more complicated intentional underreporting brownwashing phenomena.

And just for the record the findings here answer Moreno Ureba's 2026 call to disentangle corporate decoupling. This paper based on neoinstitutional stakeholder agency gender socialization and resource dependence theories analyses respectively interaction and antecedent effects and finds that the coefficients do not significantly differ from zero.

In practical terms this means that the studied mechanisms act through different paths rather than a single system of reinforcement.

We repeat the external standardization and internal governance mechanisms are different things.

The extensive versus intensive margin analysis extends neoinstitutional theory further by showing that standard adoption works as a field level constraint that prevents brownwashing and with a similarly sized but less precisely estimated coefficient corrects it once underway Meyer Rowan 1977 Roszkowska Menkes et al 2024 Wijen 2014.

Our findings carry implications for regulators directors and capital market participants Freeman 1984.

For policymakers mandating disclosure or encouraging voluntary GRI ado

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.